Daily Calculus & Graphs Drill #1 — Annotate Edition

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Derivative Notation · The Power Rule · Tangents & Normals · The Second Derivative.

Section A Rapid Recognition

1. Find the gradient of the tangent to $y = x^2 - 3x$ at $x = 2$.

Ans: 1

2. $f(x) = (x + 1)(x - 3)$. Find $f'(x)$.

Ans: $2x - 2$

3. $f(x) = \sqrt{x}$. Find $f'(4)$.

Ans: $\frac{1}{4}$

4. $f(x) = \dfrac{1}{x^2}$. Find $f'(x)$.

Ans: $-\frac{2}{x^3}$

5. $f(x) = x^4 - 2x^2$. Find $f''(1)$.

Ans: 8

6. $f(x) = x^3 - 3x^2 + 4x$. Find $f''(2)$.

Ans: 6

7. A particle moves along a line with displacement $s = t^3 - 6t^2 + 9t$ metres at time t seconds. Find its velocity at $t = 2$ and state what the sign tells you.

Ans: -3 m/s; moving in the negative direction

8. Find the equation of the normal to $y = x^2$ at $x = 1$.

Ans: $y = -\frac{1}{2}x + \frac{3}{2}$

9. $f(x) = \dfrac{x^2+4}{\sqrt{x}}$. Find $f'(4)$.

Ans: $\frac{11}{4}$

10. $f(x) = ax^3 + bx$, where $f'(1) = 5$ and $f''(1) = 12$. Find a and b.

Ans: $a = 2,\ b = -1$

Section B Manipulation Drills

1. Find the equation of the tangent to $y = x^2 - 4x + 5$ at $x = 3$.

A) $y = 2x + 8$

B) $y = -\dfrac{1}{2}x + \dfrac{7}{2}$

C) $y = 2x - 4$

D) $y = 6x - 16$

E) $y = 2x + 5$

Ans: C) $y = 2x - 4$

2. Find the equation of the normal to $y = x^2 - 2x$ at $x = 3$.

A) $y = 4x - 9$

B) $y = \frac{1}{4}x + \frac{9}{4}$

C) $y = -4x + 15$

D) $y = -\frac{1}{4}x - \frac{9}{4}$

E) $y = -\frac{1}{4}x + \frac{15}{4}$

Ans: E) $y = -\frac{1}{4}x + \frac{15}{4}$

3. For $y = x^2 + 2x$, what is the average rate of change between $x = 1$ and $x = 4$?

A) 21

B) 7

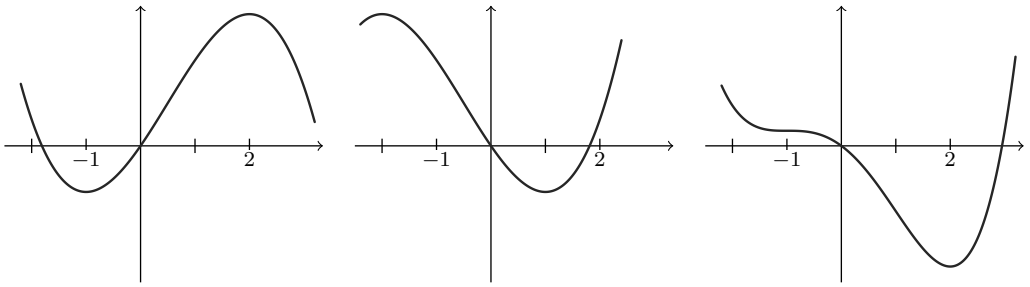
C) 4

D) 10

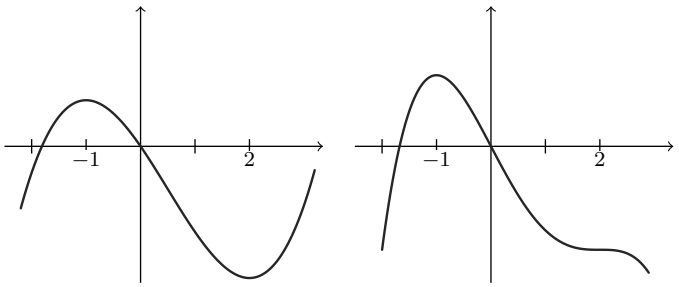
E) $\frac{21}{4}$

Ans: B) 7

4. The derivative of f satisfies $f'(x) > 0$ for $x < -1$, $f'(-1) = 0$, $f'(x) < 0$ for $-1 < x < 2$, $f'(2) = 0$, and $f'(x) > 0$ for $x > 2$. Which sketch could be $y = f(x)$?



A **B** **C**



D **E**

Ans: D

5. The tangent to $y = x^2$ at $x = a$ passes through $(3, 0)$. Find every possible value of a.

A) $a = 0$ or $a = 6$

B) $a = 6$ only

C) $a = 0$ only

D) $a = 3$

E) $a = -6$ or $a = 0$

Ans: A) $a = 0$ or $a = 6$

6. The normal to $y = x^2$ at $x = a$ is parallel to $y = -\dfrac{x}{4} + 1$. Find a.

A) $a = -2$

B) $a = 2$

C) $a = 4$

D) $a = \dfrac{1}{2}$

E) $a = \dfrac{1}{8}$

Ans: B) $a = 2$

7. The tangent to $y = \sqrt{x}$ at $x = 4$ meets the x-axis at

A) $x = 4$

B) $x = 1$

C) $x = -12$

D) $x = -4$

E) $x = \dfrac{9}{2}$

Ans: D) $x = -4$

8. For $y = x^3$, find the average rate of change between $x = 1$ and $x = 1 + h$. Simplify your answer, then deduce the instantaneous rate of change at $x = 1$.

Ans: average rate $= 3 + 3h + h^2$; instantaneous rate $= 3$

9. Find the equation of the tangent to $y = x^3 - 3x$ at $x = 2$, and find where that tangent meets the curve again.

Ans: $y = 9x - 16$; again at $(-4, -52)$

10. The tangent to $y = x^2 - 2x$ at P is parallel to $y = 2x + 5$. Find the coordinates of P, and the equation of the normal at P.

Ans: $P = (2, 0)$; normal $y = -\frac{1}{2}x + 1$

Section C Substitution & Structure

1. $f(x) = \dfrac{x^3 - 8}{x - 2}$ for $x \neq 2$. Find $f'(3)$.

A) 27

B) 8

C) 4

D) 6

E) 19

Ans: B) 8

2. The graph of $y = f'(x)$ is a straight line with positive gradient passing through $(1, 0)$. Which could be $f(x)$?

A) $x^2 + 2x - 3$

B) $-x^2 + 2x$

C) $x^3 - 3x$

D) $x^2 - 2x + 5$

E) $2x - 2$

Ans: D) $x^2 - 2x + 5$

3. A cubic $y = f(x)$ has roots $x = -1$, 2 and 4, and y-intercept -16. Find $f'(0)$.

A) -4

B) 2

C) 4

D) -10

E) -16

Ans: A) -4

4. For $f(x) = x^3 - 6x^2 + 9x$, for which x are **both** $f'(x) < 0$ and $f''(x) > 0$?

A) $1 < x < 3$

B) $1 < x < 2$

C) $x > 2$

D) $x > 3$

E) $2 < x < 3$

Ans: E) $2 < x < 3$

5. The tangents to $y = x^3 + kx^2$ at $x = 1$ and $x = 3$ are parallel. Find k.

A) 6

B) -12

C) -6

D) $-\dfrac{13}{4}$

E) No such k exists

Ans: C) -6

6. The line $y = 4x + c$ is a tangent to $y = x^3 + x$. Find all possible values of c.

A) $c = \pm 1$

B) $c = \pm 2$

C) $c = -2$ only

D) $c = \pm 3$

E) $c = \pm\dfrac{10\sqrt{3}}{9}$

Ans: B) $c = \mp 2$

7. The tangent to $y = \dfrac{4}{x}$ at $x = 3$ meets the axes at P and Q. Find the area of triangle OPQ.

A) 16

B) 0

C) 4

D) 8

E) $\dfrac{25}{2}$

Ans: D) 8

8. A cubic $f(x) = ax^3 + bx^2 + cx + d$ has tangent $y = 2x + 1$ at $x = 1$, with $f''(1) = 4$ and $f''(0) = -2$. Find $f(2)$.

A) 8

B) 5

C) 6

D) 7

E) Cannot be determined

Ans: A) 8

Section D Challenge

1. The chord of $y = x^3$ from $x = -1$ to $x = 2$ is parallel to the tangent at $x = c$, where $-1 < c < 2$. Find all such c.

A) $c = \dfrac{1}{2}$

B) $c = -1$ and $c = 1$

C) $c = \sqrt{3}$

D) $c = 1$ only

E) $c = \dfrac{2}{\sqrt{3}}$

Ans: D) $c = 1$ only

2. The curves $y = x^2$ and $y = -(x-2)^2$ have $y = 0$ as a common tangent. Find all their other common tangents.

A) $y = 4x - 4$ only

B) $y = -4x - 4$ only

C) $y = 4x - 4$ and $y = -4x - 4$

D) $y = 2x - 1$

E) There are no others

Ans: A) $y = 4x - 4$ only

3. The curve $y = f(x) = x^3 + bx^2 + cx + d$ has horizontal tangents at $x = -1$ and $x = 3$. Find $f(-1) - f(3)$.

A) -8

B) 64

C) -32

D) Cannot be determined without d

E) 32

Ans: E) 32

4. Several tangents to $y = x^3 - 3x$ pass through the point $(2, -2)$. Find the sum of the x-coordinates of all their points of contact.

A) 1

B) 3

C) 2

D) $2 + \sqrt{3}$

E) -3

Ans: B) 3

5. Let $a \neq 0$.

(a) Show that the tangent to $y = x^3$ at $x = a$ meets the curve again at $x = -2a$.

(b) Show that the gradient of the curve at $x = -2a$ is four times the gradient at $x = a$.

(c) The tangent at $x = -2a$ meets the curve again at $x = b$. Find b, and the gradient there in terms of a.

(d) Explain why the condition $a \neq 0$ is needed.

Ans: $b = 4a$; gradient $48a^2$